The LoRaWAN

ATSAWIN CHAIBAL

PHUBET SANGEAMNGAM

BACHELOR OF ENGINEERING

IN

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SCHOOL OF INFORMATION TECHNOLOGY

MAE FAH LUANG UNIVERSITY

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THIS SENIO PROJECT HAS BEEN APPROVED

TO BE A PARTIAL FULFILLLMENT OF THE REQUIREMENTS

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IN COMPUTER ENGINEERING

2020

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ABSTRACT

Nowadays, communication technology is an unavoidable technology and there are many communication technologies available, but one technology called LoRa is a low-power communication technology that can create a communication system without Using Bluetooth or Wi-Fi to communicate, it can communicate up to several kilometers. In our LoRaWan project, we tested LoRa functionality by the TTGO and ESP32 boards, sending and receiving data to each other by experimenting with the distance and frequency of the signal.

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CHAPTER 1
INTRODUCTION

1.1 Background and Rationale

The background and rationale of this project is the introduction of an old project that the university's seniors had made, LoRaWAN, a low power communication signal technology. But that project is just a prototype. Therefore, our group has reworked and developed this project to make this project more complete and to be a knowledge base for the juniors in the university as well.

1.2 Objective

The objective is to study the LoRaWAN transmission.

1.3 Scope

1.3.1 The transmission data can be transmitted to the transceiver precisely and completely.

1.3.2 The transmission scope of this project is transmission in a LoRa network using LoRaWAN.

1.4 Related Work

1.4.1 NB-IoT (Narrow Band IoT) designed by 3GPP with 3G, 4G / LTE side-spec specs, a cheap warning first announced February 2015, coming out with Release 13 of 3G 3GPP. The midpoint of the NB-IoT is that low-rate data types can be used the same network as the Blue Lar network and require little aesthetic.

The NB-IoT is designed for the following reasons.

- Long battery life: 10 years, 5-watt battery life
- 1 Cell site supports approximately 50,000 nodes.

1.5 Expected Result

The expected result of this project the transmission of information can be transmitted completely.

1.6 Methodology

We will collect all the information about the initial LoRa implementation, conduct experiments in different situations, determine the optimum values, and the conclusion of the project is that the most effective experimental results are needed.

1.7 Project Plan

Table 1.1: Project planning

Task/Mount	September	October	November	December
Finding the topic and the advisor				
Research the information				
Start to make project on software mode				
Write document chapter 1-3				
Prepare for final presentation				

Table 1.2: Project planning (cont.)

Task/Mount	JANUARY	FEBRUARY	MARCH	APRIL-MAY
Check and solve problem				
Pre-final presentation				
Documentation / Report				
Final presentation				

1.8 Resources

1.8.1 Equipment

- TTGO LoRa32 SX1276 OLED
- ESP32
- Laptop

1.8.2 Place

- S7-A Building
- S7-B Building
- Library
- Lab S3 Building

1.8.3 Budget

- TTGO LoRa32 SX1276 OLED & ESP32 x2 = 2,050.00 THB

CHAPTER 2
LITERRATURE REVIEW

2.1 Arduino

Figure 2.1 : Arduino Logo

Arduino is an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices.

2.2 TTGO LoRa32 SX1276 OLED Board

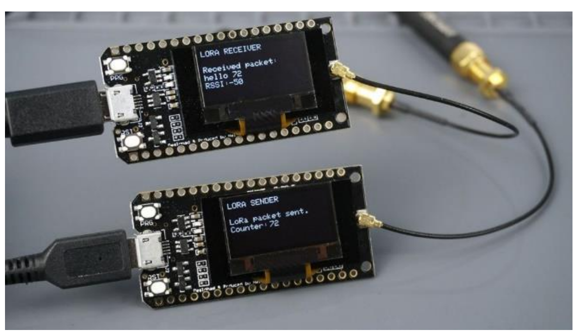

Figure 2.2 : TTGO LoRa32 SX1276 OLED Board

The TTGO LoRa32 SX1276 OLED is an ESP32 development board with a built-in LoRa chip and an SSD1306 0.96-inch OLED display. In this guide, we'll show you how to: send and receive LoRa packets (point to point communication) and use the OLED display with Arduino IDE.

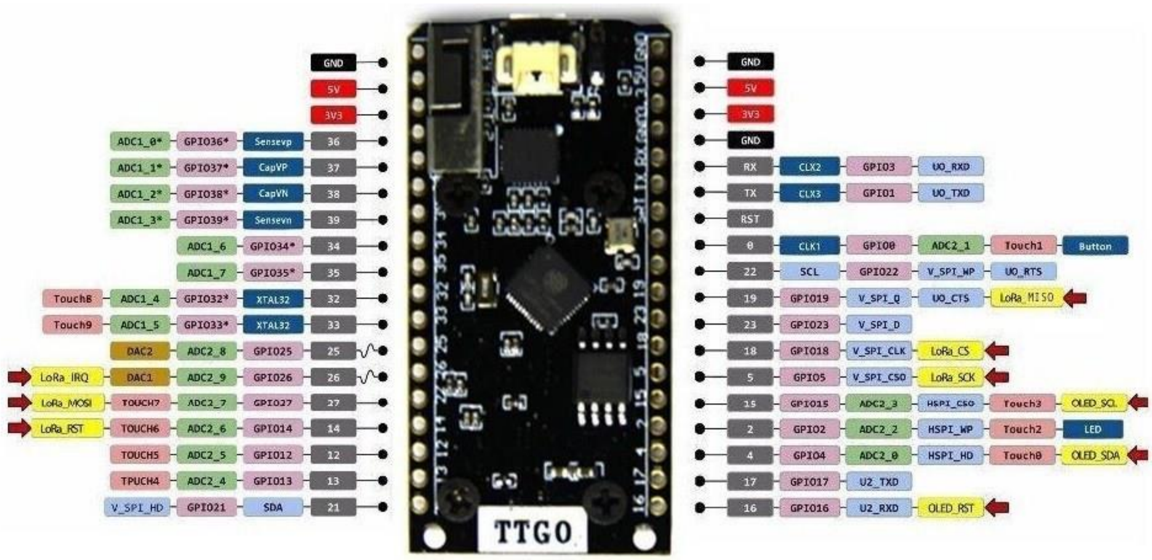

Figure 2.3 : TTGO Structure

OLED (built-in)	**ESP32**
SDA	GPIO 4
SCL	GPIO 15
RST	GPIO 16

SX1276 LoRa	**ESP32**
MISO	GPIO 19
MOSI	GPIO 27
SCK	GPIO 5
CS	GPIO 18
IRQ	GPIO 26
RST	GPIO 14

Figure 2.4 : TTGO Structure

2.3 What is LoRa & LoRaWAN

Figure 2.5 : LoRa & LoRaWAN Logo

Lora is wireless data communication technology to uses a radio modulation by Sentech LoRa transceiver chips

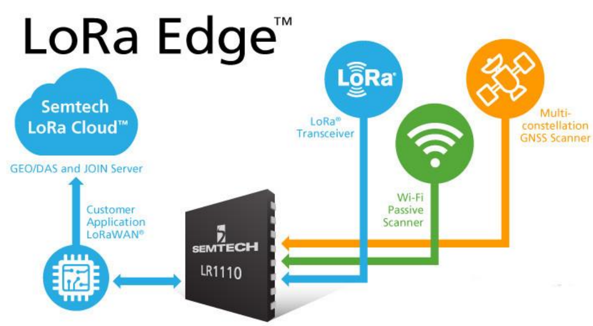

Figure 2.6 : LoRa and Sentech LoRa

This technique is also a technique that allows small amounts of information to be communicated at a distance, and to the liver against interference, in which minimization of its utilization should not be helped. Using low quality

2.3.1 LoRa frequency

The key elements of LoRa communication technology are as follows.

Long distance communication 15 - 20 kilometers, there are millions of channels for receiving signals. It has a very long battery life of up to 10 years.

LoRa uses a variety of frequencies in each area as follows.

Thailand	920 – 925 MHz
Europe	867 – 869 MHz
America	902 – 928 MHz
China	470 – 510 MHz
Korea	920 – 925 MHz
Japan	920 – 925 MHz
India	865 – 870 MHz

2.3.2 The premier app for LoRa.

LoRa's long-distance, low-power features are ideal for battery-powered sensors and low-power applications such as IoT, smart home, smart farm. And much more, so LoRa is a good choice for senser nodes that run on cells, coils or solar energy that transmit small amounts of data. LoRa is not suitable for projects that require high data rates So, LoRa is suitable for applications for less data transmission and in open areas such as smart farms at home, receiving data from a signal. The Farmhouse is delivered to the house to receive information about the farm, and one of the more popular games of the moment can also receive information with the devices in the house, for example, the door when we are home alone and someone enters the house or sensors installed at a private bitcoin mine in the house as well.

2.3.3 Point to point inspection operation

In a LoRa enabled point-to-point detection, two devices will chat using RF signals. For example, two ESP32s equipped with LoRa transceivers and transceivers, for example, are far apart or in a condition without Bluetooth or Wi-Fi signals. And you can also configure your ESP32. Don't be fooled by the LoRa Qian for a stupendous nudity at distances greater than 200 meters And the better results will depend on your LoRa setup and setup environment. There are also many other LoRa solutions that transmit over 10Km.

2.3.4 LoRaWAN Network Connection

You can create a LoRa network using LoRaWAN to connect to a multi-device network. The LoRaWAN protocol is a low power Wide Area Network (LPWAN) specification derived from LoRa technology standardized by the LoRa Alliance. The benefits of LoRa in home automation projects Similar to using a device term to measure the humidity on your farm, the farm is far from your home and where the farm doesn't have Wi-Fi , you can build up an esp32 and humidity meter, which reads and sends the data. Let us get as defined in esp32

2.3.5 Testing LoRa with ESP32

The device was tested 2x TTGO LoRa32 SX1276 OLED 915 MHz Instead of using an esp32 and a separate LoRa transceiver, there is an esp32 development board with a built-in LoRa chip and OLED that makes wiring easier.

2.4 LoRa vs LoRaWan

Telecommunication service providers in our home during this period, both AIS (AIS NB-IoT) and True (True IoT) have launched NB-IoT services.CAT comes via LoRa cable (LoRa IoT by CAT), both of which Was released to support communication for IoT services in the part of AIS and True who are already mobile service providers Is there a frequency allocated for LTE service, the NB-IoT servicing It will save a lot of costs, plus it is to Utilize the frequency that itself is allowed to come. As you can see, these are the 3 major service providers that have been launched. In today's topic we will discuss LoRa and Mesh Network sharing, so let's look at the differences between LoRa and LoRaWan first.

2.4.1 LoRa

Implements a physical layer module, which is suitable for P2P applications, without encryption, broadcast data, and if someone sets up a LoRa node and uses the same settings as ours, it can be seen. Data, which LoRa was developed as Wireless Data Communication by Cycleo (Grenoble, France), then the manufacturer Sentech bought this company in 2012 if you look at the specs of the hit SX1276 at home, the area 920-925MHz according to the announcement of NBTC will see that LoRa can only transmit that distance, partly because of its high sensitivity of -148dBm.

2.4.2 LoRaWan

On the top of the LoRa Radio movement, will also include encryption protocols that use AES-128 to perform QOS. LoRaWan details include: Is far enough, including the Device End Node, it can still be studied into Class ABC again, and the work will be divided into levels.

It is appropriate to choose LoRa or LoRaWan, it should be appropriate for the usage that will be used as well, and more importantly, it should be in accordance with the regulations of the NBTC since the system is used together if using LoRa. You have to look at the duty cycle itself, too, do not send information all the time, reserve the use of the system throughout, including the sending power that if too much will interfere with the use of the device or the application used in the system.

There are some drawbacks to using LoRaWan, regardless of how good technology LoRa / LoRaWan is, the occipital mate for every application, I rarely mesh Network, it will not be suitable for certain applications or applications, because if we If you want to get LoRa / LoRaWan to transmit data in real time, it would be unpredictable, it would be inconvenient to reserve tickets at all times, or to send large data would also be inconsistent, because it was limited by a few bytes of transmission size and speed. Low transmission

Suitable use-cases for LoRaWAN:

- Long range – multiple kilometers
- Low power – can last months (or even years) on a battery
- Low cost – less than 20€ CAPEX per node, almost no OPEX
- Low bandwidth – something like 400 bytes per hour
- Coverage everywhere – you are the network! Just install your own gateways
- Secure – 128bit end-to-end encrypted
- Geolocation / Triangulation – you should probably use GPS for this, but we're doing our best to make it work with just LoRa. Check out Collops.

Not Suitable for LoRaWAN:

- Realtime data – you can only send small packets every couple of minutes
- Phone calls – you can do that with GPRS/3G/LTE
- Controlling lights in your house – check out ZigBee or Bluetooth
- Sending photos, watching Netflix – check out WIFI
-

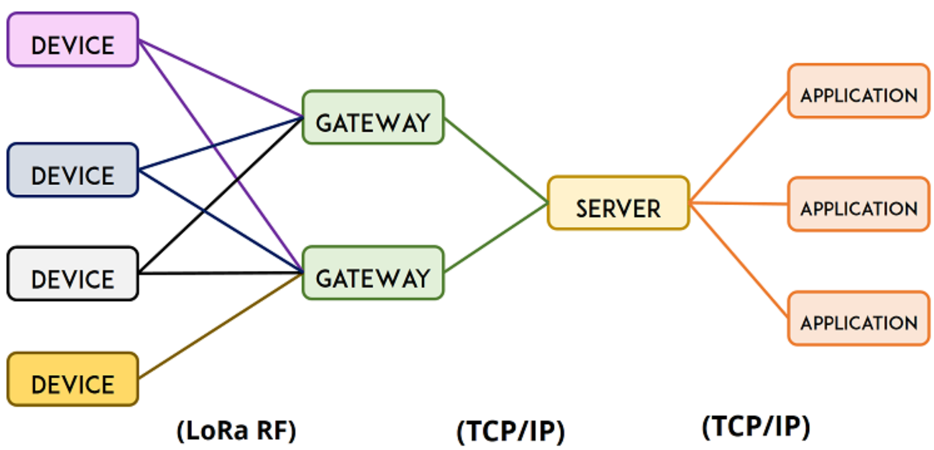

Figure 2.7 : LoRa & LoRaWAN Device to Gateway

CHAPTER 3

ANALYSIS AND PERFORMANCE TEST

3.1 Analysis and Testing LoRa with ESP32

The device was tested 2x TTGO LoRa32 SX1276 OLED 915 MHz Instead of using an ESP32 and a separate LoRa transceiver, there is an ESP32 development board with a built-in LoRa chip and OLED that makes wiring easier. Arduino IDE has an add-on for the Arduino IDE that allows you to install ESP32 programs using Arduino IDE.

Step 1 Install LoRa Library.

There are several libraries that can send and receive LoRa packages with a lot of ESP32. Open your Arduino IDE and go to Sketch> Include Library> Manage Libraries and search for “LoRa “. And choose to install as shown.

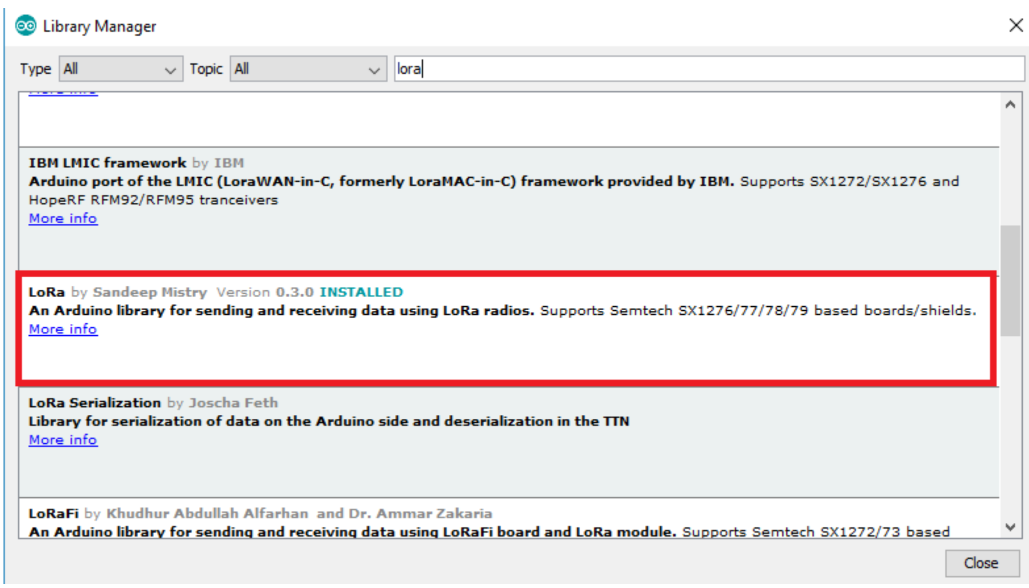

Figure 3.1 : Arduino IDE Library

Receiving LoRa Transceiver Modules

LoRa Message Sending and Receiving With ESP32, we will use the RFM95 module by transmitting and receiving, all LoRa modules are transceiver, which means to be able to send and receive signals, two or more devices are required. You can also use other compatible modules, such as Sentech SX1276 / 77/78/79 bauds, RFM96W, RFM98W, etc. There is also an eps32 baud display on an OLED display like TTGO LoRa32.

Figure 3.2 : LoRa SX1276

Figure 3.3 : TTGO LoRa SX1278

After preparing the equipment, let's move on to the process of entering code for receiving and sending signals in the Arduino IDE program.

LoRa Simple Code in Arduino IDE

LoRa Dump Registers

LoRa register dump examples to shows how to inspect and output on the LoRa radio’s registers on the Serial interface

Example code

```
#include <SPI.h
#include <LoRa.h>

void setup() {
  Serial.begin(9600); // initialize serial
  while (!Serial);

  Serial.println(“LoRa Dump Registers”);

  // override the default CS, reset, and IRQ pins (optional)
  // LoRa.setPins(7, 6, 1); // set CS, reset, IRQ pin

  if (!LoRa.begin(915E6)) {
    Serial.println(“LoRa init failed. Check your connections.”);
    while (true);
// if failed, do nothing
  }
  LoRa.dumpRegisters(Serial);
}
void loop() {
}
```

LoRa Set Spread

The spreading factor can be set from SF7-SF12, which, if we want to use with high bandwidth, must be set to SF7, which will make the transmission distance low. But the data transmission rate is high (Bit rate), but if we want to transmit data for a long distance, it must be set to SF12, which will affect the transmission rate as little as possible. But get the longest distance By setting the parameters depending on the needs of the system designer.

Example Code

```
#include <SPI.h>              // include libraries
#include <LoRa.h>

const int csPin = 7;          // LoRa radio chip select
const int resetPin = 6;       // LoRa radio reset
const int irqPin = 1;         // change for your board; must be a hardware interrupt pin

byte msgCount = 0;            // count of outgoing messages
int interval = 2000;          // interval between sends
long lastSendTime = 0;        // time of last packet send

void setup() {
  Serial.begin(9600);                   // initialize serial
  while (!Serial);

  Serial.println("LoRa Duplex – Set spreading factor");

  // override the default CS, reset, and IRQ pins (optional)
  LoRa.setPins(csPin, resetPin, irqPin); // set CS, reset, IRQ pin

  if (!LoRa.begin(915E6)) {             // initialize ratio at 915 MHz
    Serial.println("LoRa init failed. Check your connections.");
    while (true);                       // if failed, do nothing
  }

  LoRa.setSpreadingFactor(8);           // ranges from 6-12,default 7 see API docs
  Serial.println("LoRa init succeeded.");
```

```
}

void loop() {
  if (millis() – lastSendTime > interval) {
    String message = “HeLoRa World! “;   // send a message
    message += msgCount;
    sendMessage(message);
    Serial.println(“Sending “ + message);
    lastSendTime = millis();            // timestamp the message
    interval = random(2000) + 1000;    // 2-3 seconds
    msgCount++;
  }

  // parse for a packet, and call onReceive with the result:
  onReceive(LoRa.parsePacket());
}

void sendMessage(String outgoing) {
  LoRa.beginPacket();                   // start packet
  LoRa.print(outgoing);                 // add payload
  LoRa.endPacket();                     // finish packet and send it
  msgCount++;                           // increment message ID
}

void onReceive(int packetSize) {
  if (packetSize == 0) return;          // if there’s no packet, return

  // read packet header bytes:
  String incoming = “”;

  while (LoRa.available()) {
    incoming += (char)LoRa.read();
  }

  Serial.println(“Message: “ + incoming);
  Serial.println(“RSSI: “ + String(LoRa.packetRssi()));
  Serial.println(“Snr: “ + String(LoRa.packetSnr()));
  Serial.println();
}
```

LoRa Sync Word

The LoRa transceiver module listens to packets that are in the range. It doesn’t matter where the packets come from. To make sure you only receive packets from your sender, you can set up a sync word (range 0 to 0xFF).

Example Code

```
LoRa.setSyncWord(0xF3);  // ranges from 0-0xFF, default 0x34, see API docs
Serial.println(“LoRa init succeeded.”);
```

LoRa Sender

Code example for Sender

```
//Libraries for LoRa ไลบรารี่เพื่อโต้ตอบกับบอดร์ของLORA
#include <SPI.h>
#include <LoRa.h>

//Libraries for OLED Display การติดต่อการแสดงผลกับ OLED Display
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

//define the pins used by the LoRa transceiver module
//ตัวกำหนดพินที่ใช้โดยตัวรับส่งสัญญาณเป็นบอดร์LORA
#define SCK 5
#define MISO 19
#define MOSI 27
#define SS 18
#define RST 14
#define DIO0 26

//เลือกความถี่ของบอดร์ LORA
//ความถี่สำหรับประเทศไทยใช้ “915E6” ของ North America
#define BAND 915E6

//OLED pins การกำหนดพินของ OLED บนบอดร์
#define OLED_SDA 4
```

```
#define OLED_SCL 15
#define OLED_RST 16
//กำหนดขนาดของ OLED
#define SCREEN_WIDTH 128 // ความกว้างในระดับในระบบพิกเซล
#define SCREEN_HEIGHT 64 // ความยาวในระดับในระดับพิกเซล

//packet counter สร้างเคาน์ตเอร์เพื่อติดตามในนวนแพ็กเกตของLORA ที่ส่ง
int counter = 0;

//สร้างไฟล์ Adafruit_SSD1306แสดงในสกรีน
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RST);

//การติดตั้ง
void setup() {

 //reset OLED display via software
 //การติดตั้งต้องประกาศพินRSTเป็นOutputให้ตั้งค่าเป็นต่ำเป็นเวลา2-3msจากนั้นตั้งค่าเป็นระดับสูง
 pinMode(OLED_RST, OUTPUT);
 digitalWrite(OLED_RST, LOW);
 delay(20);
 digitalWrite(OLED_RST, HIGH);

 //initialize OLED
 //เริ่มการสื่อสารกับ I2Cโดยใช้ไฟล์OLED_SDAและOLED_SLCและหมุนที่ใช้เป็นWire.begin()
 Wire.begin(OLED_SDA, OLED_SCL);

 //หลังจากนั้นก็เริ่มต้นการแสดงผลด้วยพารามิเตอร์ให้ตรวจสอบให้แน่ในว่าไลบรารี่ไม่ได้ใช้พินI2C
 //โดยกำหนดพินที่ใช้ในโค้ดGPIP4,GPIO15
 if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3c, false, false)) {
  // Address 0x3C for 128x32
  Serial.println(F("SSD1306 allocation failed"));
  for( // Don't proceed, loop forever
 }

 //เขียนข้อความการแสดงผลบนหน้าจอ
 display.clearDisplay();
```

```
  display.setTextColor(WHITE);
  display.setTextSize(1);
  display.setCursor(0,0);
  display.print("AON SENDER "); //ตัวแสดงผลบนหน้าจอ
  display.display();


  //initialize Serial Monitor
  //มอนิเตอร์แบบอนุกรมเพื่อการตีบัค
  Serial.begin(115200);
  Serial.println("LoRa Sender Test");

  //SPI LoRa pins
  //กำหนดพินSPIที่บอดร์LORA
  SPI.begin(SCK, MISO, MOSI, SS);

  //setup LoRa transceiver module
  //การตั้งค่าโมดูลตัวรับส่งสัญญาณLORA
  LoRa.setPins(SS, RST, DIO0);

  //โมดูลตัวรับส่วสัญญาณLORAโดยใช้LORAส่งผ่านความถี่เป็นอาร์กิวเมนต์
  if (!LoRa.begin(BAND)) {
    Serial.println("Starting LoRa failed!");
    while (1);
  }

  //เริ่มต้นการแสดงผลตอนเปิดบอร์ด
  Serial.println("LoRa Initializing OK! START");
  display.setCursor(0,10);
  display.print("LoRa Initializing OK!");
  display.display();
  delay(2000);
}

void loop() {
  //ลูป
  Serial.print("Sending packet: ");
```

```
  Serial.println(counter);

  //Send LoRa packet to receiver
  //ส่งไปถึงตัวรับ
  LoRa.beginPacket();
  LoRa.print("PJMK ");
  LoRa.print(counter);
  LoRa.endPacket(); //ตัวปิดแพ็กเกต

  //เขียนไฟล์เคาร์เตอร์บนจอแสดงผลOLED
  display.clearDisplay();
  display.setCursor(0,0);
  display.println("AON SENDER");
  display.setCursor(0,10);
  display.setTextSize(1);
  display.print("LoRa packet sent.");
  display.setCursor(0,20);
  display.print("Counter Packet:");
  display.setCursor(100,20);
  display.print(counter);
  display.display();

  counter++;

  delay(1000);
}
```

Explain in the code part

Start by merging the necessary libraries.

```
#include <SPI.h>
#include <LoRa.h>
```

The right pin is then assigned by LoRa's module, but if you have a superior ESP32 with LoRa built-in, verify that is used by the LoRa module in your North York and determine the correct one.

```
#define ss 5
#define rst 14
#define dio0 2
```

Start the duck flea bait that starts with 0;

```
int counter = 0;
```

In the setup(), you initialize a serial communication

```
Serial.begin(115200);
while (!Serial);
```

Set the pins for the LoRa module.

```
LoRa.setPins(ss, rst, dio0);
```

Start the transceiver and transmit module with the frequency specified, the frequency will change according to your region. For example, Thailand allows the use of frequency 915.

```
While (!LoRa.begin(915E6)) {
  Serial.println(".");
  delay(500);
}
```

The LoRa transceiver module listens to packages in the range. It is not more important than any package. To make sure that only your salvaged packages are received, you can set it up for sync (0 to 0xFF).

```
LoRa.setSyncWord(0xF3);
```

Both recipient and sender need to have the same sync command.

```
LoRa.beginPacket();
```

Writes the information to the package using the print () file.

```
LoRa.print("hello ");
LoRa.print(counter);
```

Then close the package with the endPacket () file.

```
LoRa.endPacket();
```

After this, the message will respond after the specified time.

```
Counter++;
delay(10000);
```

Tests for Sender

Upload the code to the ESP32 baud and make sure you have selected the correct baud and COM port, then open Serial Monitor and press the enable button, ESP32 the message will be sent as seen. In the picture

LoRa Receiver
Code for LoRa receiver

Now another ESP32 uploaded this sketch will come with an R RSSI designation, measuring the strength of both received and transmitted signals.

```
#include <SPI.h>
#include <LoRa.h>

//Libraries for OLED Display
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

//define the pins used by the LoRa transceiver module
#define SCK 5
#define MISO 19
#define MOSI 27
#define SS 18
#define RST 14
#define DIO0 26

//433E6 for Asia
//866E6 for Europe
//915E6 for North America
#define BAND 915E6

//OLED pins
#define OLED_SDA 4
#define OLED_SCL 15
#define OLED_RST 16
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RST);

String LoRaData;
```

```
void setup() {

  //reset OLED display via software
  pinMode(OLED_RST, OUTPUT);
  digitalWrite(OLED_RST, LOW);
  delay(20);
  digitalWrite(OLED_RST, HIGH);

  //initialize OLED
  Wire.begin(OLED_SDA, OLED_SCL);
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3c, false, false)) { // Address 0x3C
for 128x32
    Serial.println(F(“SSD1306 allocation failed”));
    for( // Don’t proceed, loop forever
  }

  display.clearDisplay();
  display.setTextColor(WHITE);
  display.setTextSize(1);
  display.setCursor(0,0);
  display.print(“LORA RECEIVER “);
  display.display();

  //initialize Serial Monitor
  Serial.begin(115200);

  Serial.println(“LoRa Receiver Test”);

  //SPI LoRa pins
  SPI.begin(SCK, MISO, MOSI, SS);
  //setup LoRa transceiver module
  LoRa.setPins(SS, RST, DIO0);

  if (!LoRa.begin(BAND)) {
    Serial.println(“Starting LoRa failed!”);
    while (1);
  }
  Serial.println(“LoRa Initializing OK!”);
  display.setCursor(0,10);
```

```
  display.println(“LoRa Initializing OK!”);
  display.display();
}

void loop() {

  //try to parse packet
  int packetSize = LoRa.parsePacket();
  if (packetSize) {
    //received a packet
    Serial.print(“Received packet “);

    //read packet
    while (LoRa.available()) {
      LoRaData = LoRa.readString();
      Serial.print(LoRaData);
    }

    //print RSSI of packet
    int rssi = LoRa.packetRssi();
    Serial.print(“ with RSSI “);
    Serial.println(rssi);

   // Dsiplay information
   display.clearDisplay();
   display.setCursor(0,0);
   display.print(“LORA RECEIVER”);
   display.setCursor(0,20);
   display.print(“Received packet:”);
   display.setCursor(0,30);
   display.print(LoRaData);
   display.setCursor(0,40);
   display.print(“RSSI:”);
   display.setCursor(30,40);
   display.print(rssi);
   display.display();
  }
}
```

Code explanation

This receiver is similar to the sender and only has a different loop () and may change the sync word frequency to match the frequency used in the sender's body by loop (). The code checks the pack. Re-use the file. parsePacket ()

```
int packetSize = LoRa.parsePacket();
```

If there is a new package, we read the incoming content using the readString () file.

```
While (LoRa.available()) {
  String LoRaData = LoRa.readString();
  Serial.print(LoRaData);
}
```

And finally, the incoming data will be saved in LoRa Data file and printed in Serial Monitor RSSI, it will be received in dB.

```
Serial.print("' with RSSI ");
Serial.println(LoRa.packetRssi());
```

Testing for LoRa Receiver

Upload the code to ESP32 and connect the two bauds to each other by updating Serial Monitor for the LoRa Receiver and pressing the Open button in the LoRa Sender to start receiving LoRa packages on the receiver.

3.2 Performance test

Test1 Cording test in Arduino in TTGO LoRa32 Board and remote test

From our first experiment, we performed a long distance test using an entry-level signal using a 3dBi antenna, tested for signals at different distances, and then looked at the RSSI level and the missing signal at a distance of 220 meters. For this experiment, we used the frequency 915.

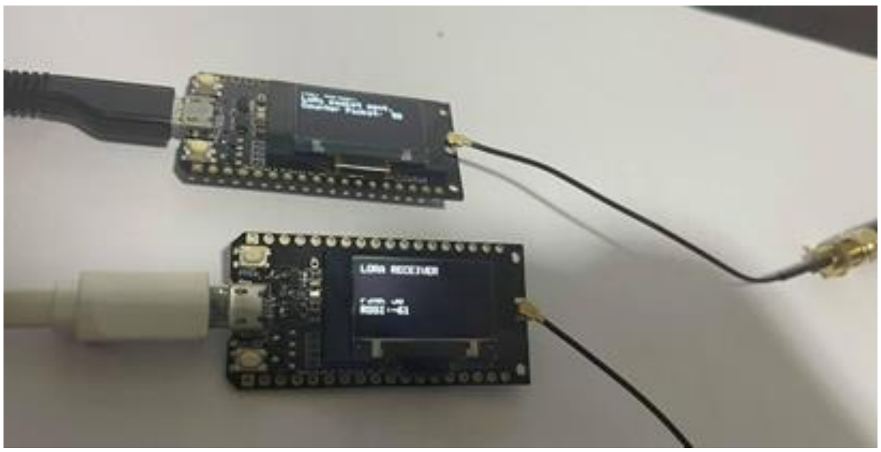

Figure 3.4 Testing Board on TTGO

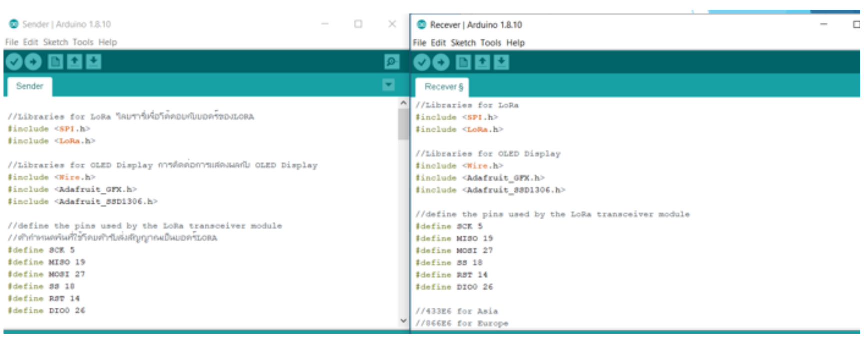

Figure 3.5 : cording test in Arduino in TTGO LoRa32 Board

Experiment 1 We conducted an experiment in an open area free of obstructions. It appears that the signal starts to fade away at a distance of 220 meters and the RSSI value is 121.

Figure 3.6 : Horizontal test area image

Experiment 2 We measured signals from different distances with the same obstructions and distances, concluding that the RSSI signal increased by 3 to 4. From 100 meters to the average distance of signal loss is 220 meters.

Figure 3.7: Horizontal test area image In areas of different levels and there are obstacles

Table 2: Horizontal signal test results

Distance	**dB**
Close together	24
Close together and remove the receiver head	54
1m	74
2m	77
3m	74
4m	89
5m	99
8m	101
11m	105
13m	114
15m	106
17m	107
19m	111
21m	113
23m	117
25m	104
27m	112
30m	116
50m	115
100m	113
150m	117
220m	119
220m+	121 And the signal is missing

Test 2 -Vertical

For this experiment, we performed this experiment in buildings with many obstacles, including routers, signals, and walls of a multi-storey building. It can be concluded that the distance has decreased significantly from the first experiment at 220 meters. But in this experiment, the maximum distance is an average of 37 meters.

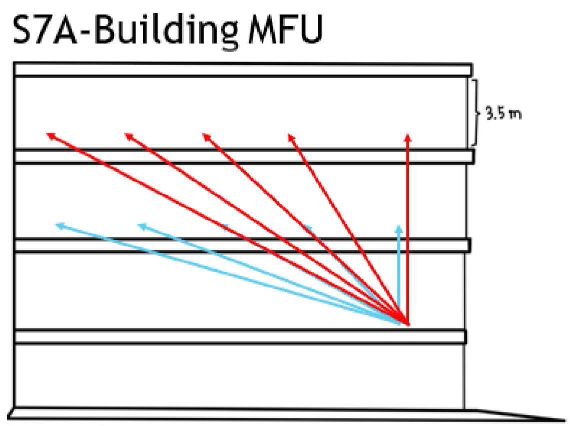

Figure 3.8: S7 Building vertical test scenario

Table 3.1: Indoor vertical signal test results

Distance	Vertical 3.5m	Oblique line9.84m	Oblique line18.72m	Oblique line27.82m	Oblique line36.96m
Between the 2nd and 3rd floors there is a ceiling separating between floors	**-104 db**	**-107db**	**-114 db**	**-123 db**	**-119 db**

Table 3.2: Indoor vertical signal test results(cont.)

Distance	Vertical 7m	Oblique line11.56m	Oblique line18.72m	Oblique line28.47m	Oblique line37.45m
Between the 2^{nd} and 4^{th} floors there is a ceiling separating between the 2 floors	**-114 db**	**-120db**	**-121 db**	**-118 db**	**-117 db**

Test 3 Test the signal level by increasing the signal directly and while moving.

The results of the distance test of each level used to test the picture above, which if taken to the longest result, which is in the support line, will be about 500 meters for receiving-transmitting technology at 923.2MHz, while the frequency is 433.175 MHz. The result will be approximately 400 meters.

Function LoRa.setTxPower (); Is a function used to determine the signal strength when transmitting data. There are patterns of use as follows
void LoRa.setTxPower (int txPower); Details of the parameters are as follows.
(int) txPower - The desired transmission unit is set in dBm. 1 to 17 can be specified.

Figure 3.9: Horizontal test area

In addition, we tested the same distance with building obstructions, concluding that the received signal level did not differ from that of the open area because the signaling area was also an open area.

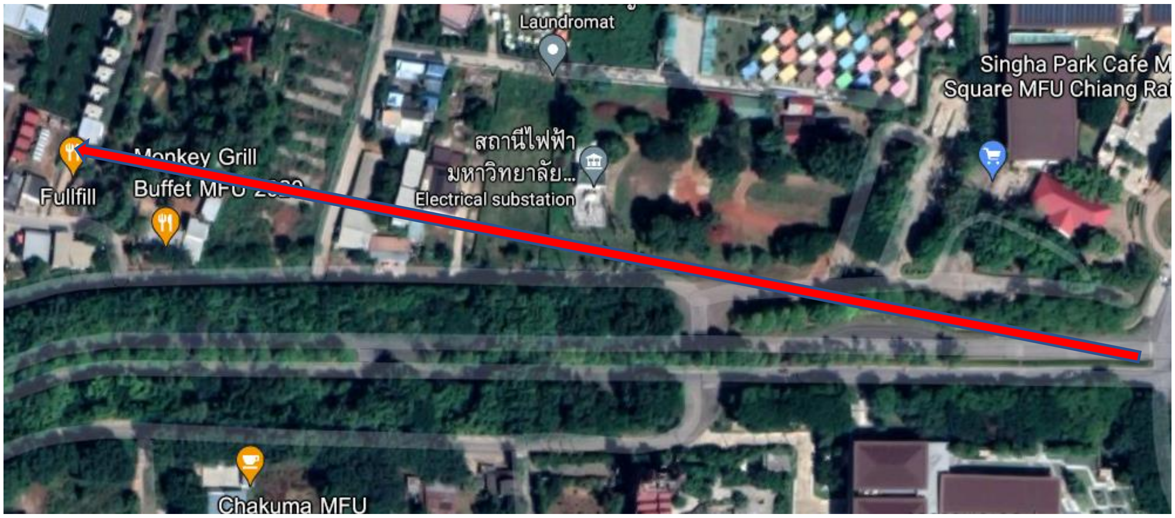

Figure 3.10: Horizontal test area by various areas and with obstacles

We then tested the signal moving at an average speed of 70 kilometers per hour and watched it vibrate at + -3 all the time, compared to not moving it at + -1.

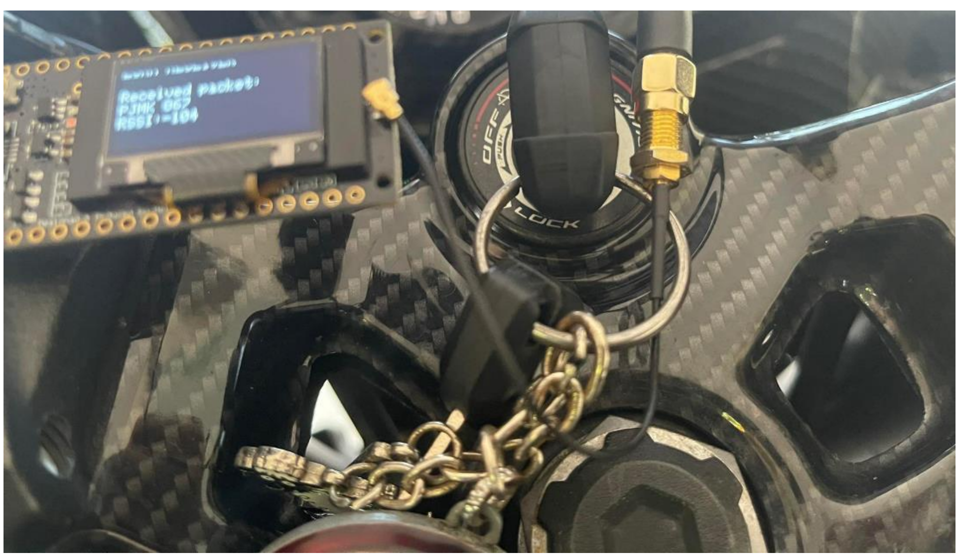

Figure 3.11: Test the signal with a movement of 70 kilometers per hour

Table 4: Horizontal signal test results

Distance	**dB**
Close together	24+-
Close together and remove the receiver head	54+-
1m	74+-
50m	88+
100m	91+-
150m	94+-
200m	104+-
300m	105+-
400m	115+-
500m	117+-
540m+	119+-

Test 4 The test was performed using different antenna heights and receiver distances

For this test, we performed a test by comparing the height of the main pole 915. 3dBi gain The different heights are located at 0 to 1 m. 3 to 4 m, 6 to 7 m, and 9 to 10 m respectively, with the Node moving away. Until the signal disappears

The results of the experiments pole setting at 0 to 1 m in the range of 200 to 220 m, as in the first experiment, the RSSI values were highest at 119 + -, the 3 to 4 m heights were in the range 250 to 280. The micrometer is slightly increased. Maximum RSSI value is 121 + -, pole height of 6 to 7 m is range of 350 to 450 m slightly increased. Maximum RSSI value is 122 + -, pole height of 9 to 10 meters. Has been in the range of 500 to 540, a slight increase, the RSSI value is 123+-.

Table 4: Test results from different altitudes and distances

Distance height	0-1m	3-4m	6-7m	9-10m
RSSI	119+-	121+-	122+-	123+-
Distance	200-220m	250-280m	350-450m	500-540m

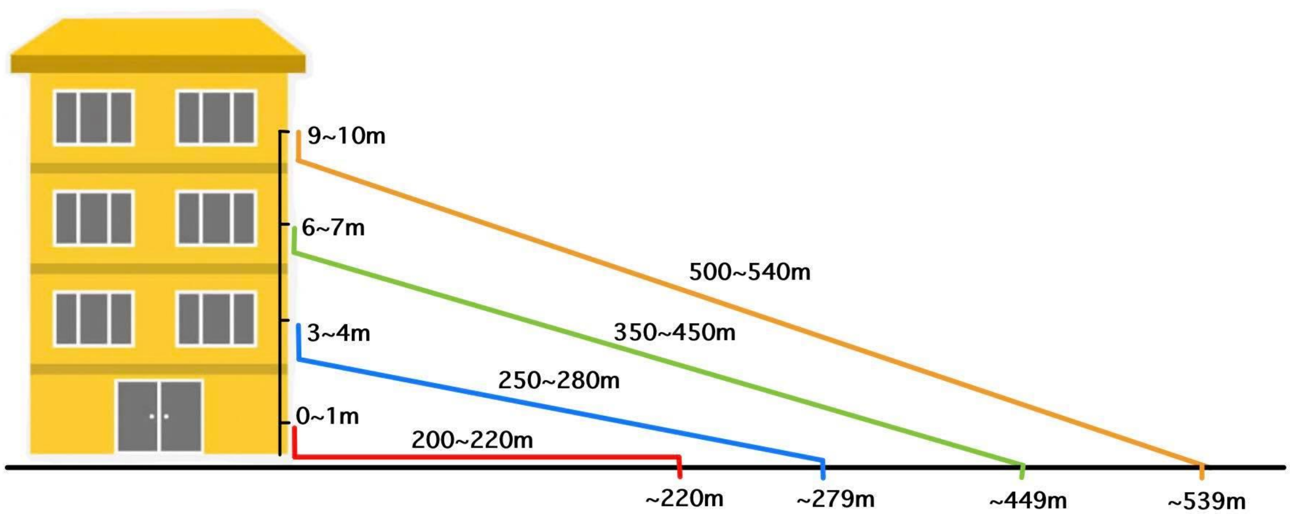

Figure 3.12 : S3 Building vertical test scenario

CHAPTER 4
CONCLUSION

4.1 Conclusion

LoRa testing in different situations and locations. By setting the pole higher and less obstructions, the LoRa transmission range increases, as well as the associated hardware and code. The maximum RSSI is 124, after which the signal is lost. And most importantly, the signal is minimized when the transmitter or receiver is in an area with many obstructions, for example an obstruction, a router that interferes with the signal, or an area with large obstructions.

4.2 Future Work

In this experiment, the project could only transmit a maximum distance of 540 meters. Using a 3dBi antenna at a height of 9 to 10 meters only. Therefore, if in the future there is a signal that is stronger than this and add Main and Node will be able to amplify the signal more.

REFERENCE

- Randomnerdtutorials (October 19, 2019) : https://randomnerdtutorials.com/ttgo-lora32-sx1276-arduino-ide/

- Getting Started: FOUNDATION > Introduction". arduino.cc.
  David Kushner (2011-10-26). "The Making of Arduino". IEEE Spectrum.

- "What is LoRa? | Semtech LoRa Technology | Semtech". www.semtech.com. Retrieved 2021-01-21.

- ist, Jenny (2020-11-22). "Espressif Leaks ESP32-C3: A WiFi SOC That's RISC-V and is ESP8266 Pin-Compatible". Hackaday.

- "LoRaWAN For Developers". www.lora-alliance.org. Retrieved 2018-11-23.